Analytical Chemistry Cumulative Exam
October 17, 2003

1. (10 points) What does the acronym SECM stand for?

2. (30 points) What is SECM good for? (What information can it obtain from samples? What kinds of samples can it analyze and under what kinds of conditions? What are some applications for SECM?)

3. (45 points) How does SECM work? (In answering this question, just pick one kind of SECM… you don't need to explain how all different kinds of SECMs work.)

(a) (25 points) Explain the main physical and chemical principles upon which SECM is based. Draw a schematic to help you in your explanation and show how the signal is obtained from the sample.

(b) (20 points) Draw a diagram of the portion of the instrument that interacts with a sample and label the parts. Draw a block diagram of the rest of the major instrumental components and label them.

4. (15 points) Explain positive feedback and negative (or blocking) feedback modes in SECM.